

FASHION COMMERCE

PROJECT: Fashion Commerce Analytics Case Study

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SALES

PRODUCT & INVENTORY

CAMPAIGN

CUSTOMER & MARKET

2025

FASHION COMMERCE

BUSINESS CONTEXT

TIVA is a premium fashion brand — average order value around \$358 — selling online across Spain, Portugal, the Netherlands, Italy, Germany, and France. Over the period it did roughly \$324,000 in revenue and about 900 orders. Growth is now coming from paid campaigns and promotions rather than pure organic demand, and management has the numbers scattered across spreadsheets with no single view.

- LIMITED VISIBILITY INTO CAMPAIGN ROI**
The brand can see campaign activity, but cannot clearly identify which campaigns generate enough revenue and profit to justify the spend.
- NO MARKET-SPECIFIC COMMERCIAL STRATEGY**
The brand applies the same commercial approach across six European markets despite meaningful differences in customer value, loyalty, and purchasing behaviour.
- HEAVY DISCOUNTING, UNCLEAR CUSTOMER VALUE**
The brand is discounting heavily without knowing whether it is attracting loyal, high-value customers or one-time bargain hunters.

DATA METHODOLOGY

DATA PROFILING	checked keys, missing values, duplicates, ranges, and table relationships
DATA CLEANING	standardized dates, discounts, prices, and categorical fields in Python
DATA VALIDATION	reconciled sales totals and verified foreign-key consistency
DATA TRANSFORMATION	created analysis-ready datasets for sales, products, customers, inventory, and campaigns
KPI ENGINEERING	calculated GMV, AOV, Gross Margin, Sell-through, ROAS, CAC, Conversion Rate, and customer metrics
CUSTOMER SEGMENTATION	built RFM-based customer groups and repeat-purchase measures
SYNTHETIC EXTENSION	generated campaign spend and funnel data using original campaign IDs, channels, and periods
BI MODELLING	connected prepared datasets to Looker Studio for interactive analysis



Sales Performance

Channel

Category

Country

12 May 2025 - 1 Jun 2025

Revenue
\$125,996
↑ 54.9%

GMV
\$132,778
↑ 63.2%

Orders
359
↑ 58.8%

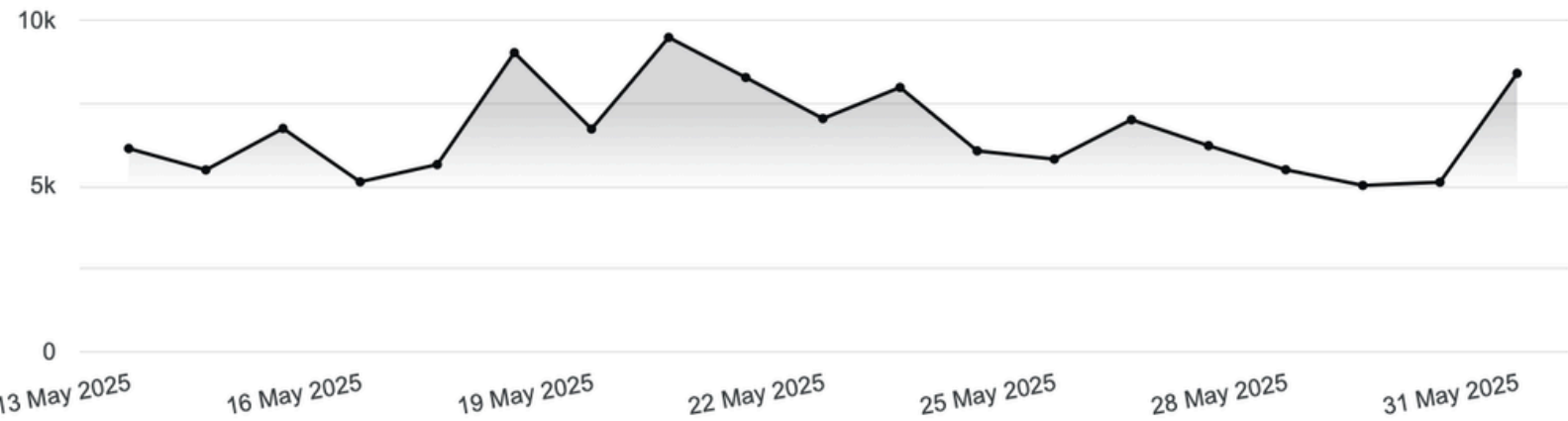
Units Sold
2,694
↑ 62.3%

AOV
\$351
↓ -2.5%

Gross Margin
42.23%
↓ -6.5%

Gross Profit
\$53,211
↑ 44.9%

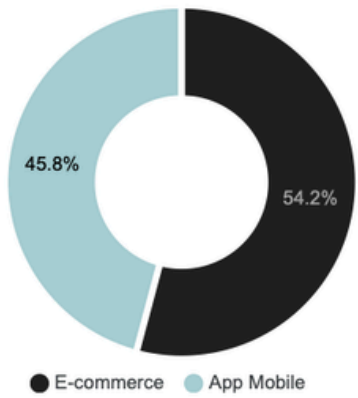
Revenue Trend



Revenue by Category



Revenue by Channel

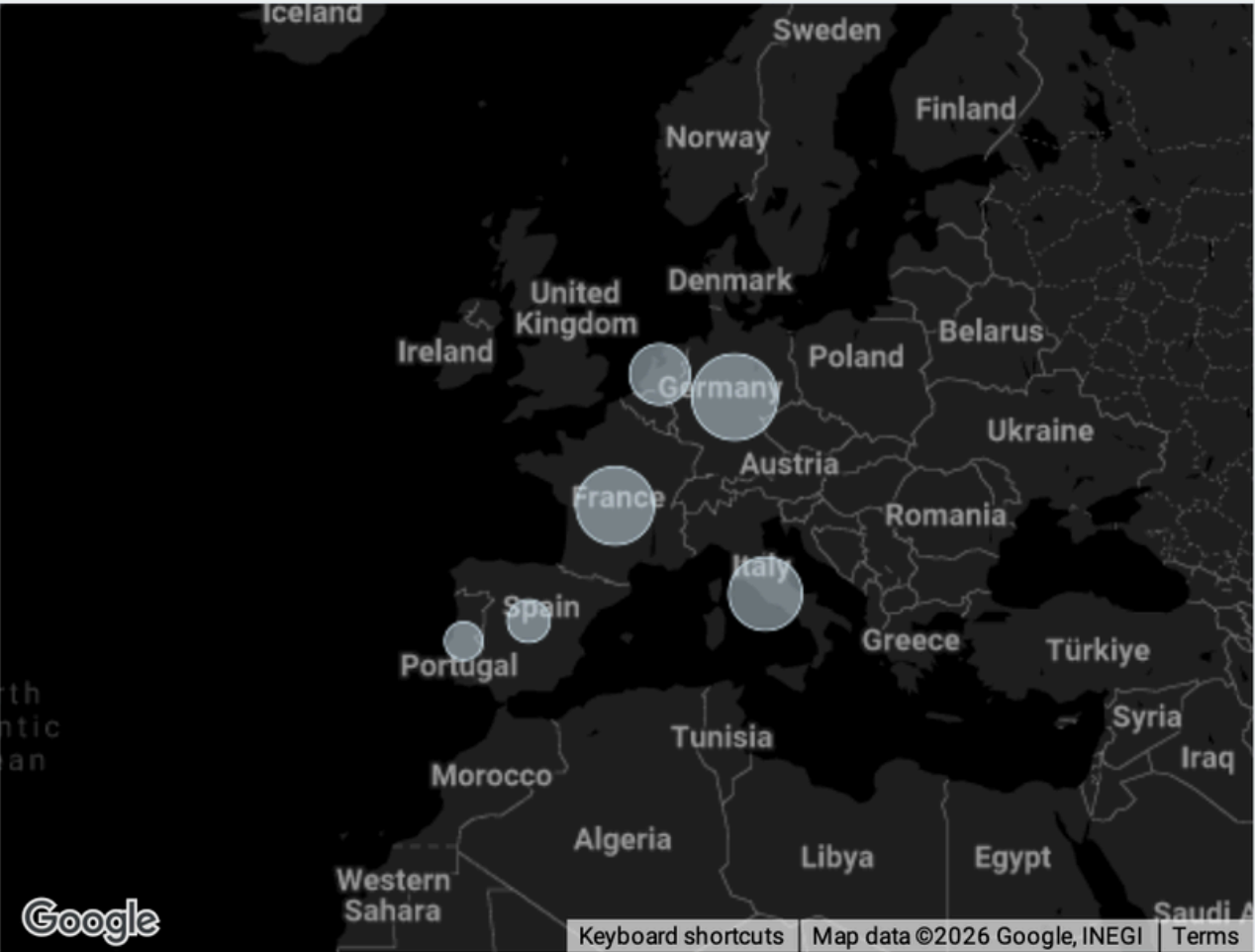


Sales Performance Table

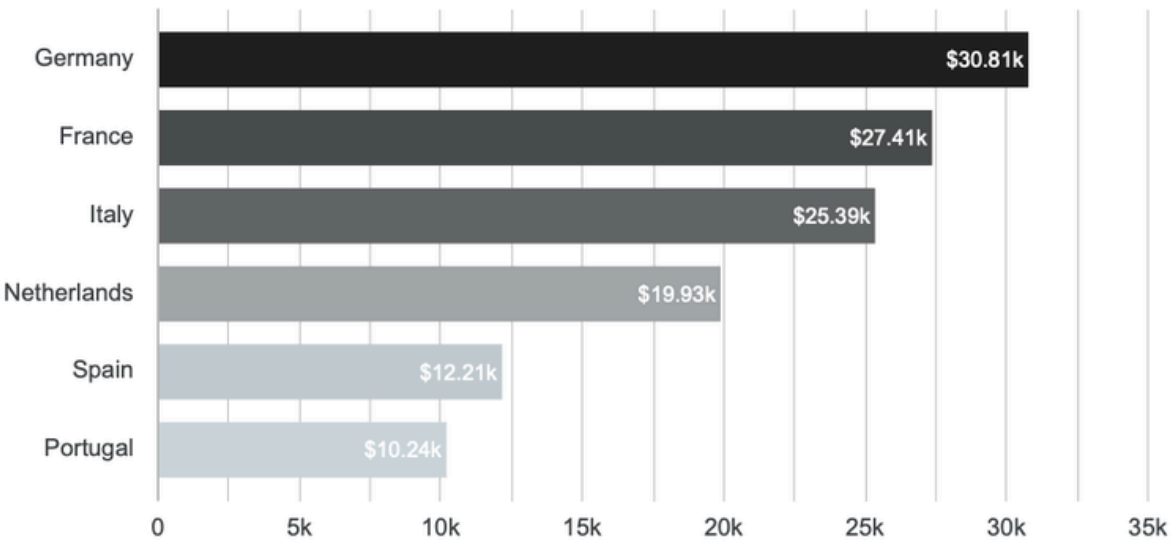
	Country	Channel	Revenue ▾	GMV	Orders	Units	AOV	Gross Margin
1.	Germany	App Mobile	\$15,938.36	\$17,671.02	43	360	\$370.66	39.43%
2.	Germany	E-commerce	\$14,874.69	\$14,874.69	42	308	\$354.16	45.99%
3.	France	E-commerce	\$14,293.53	\$14,293.53	39	289	\$366.5	43.72%
4.	Italy	E-commerce	\$13,385.53	\$13,385.53	37	271	\$361.77	45.95%
5.	France	App Mobile	\$13,116.64	\$14,151.46	40	287	\$327.92	39.41%
6.	Italy	App Mobile	\$12,008.88	\$13,594.42	37	273	\$324.56	35.83%

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Revenue by Country



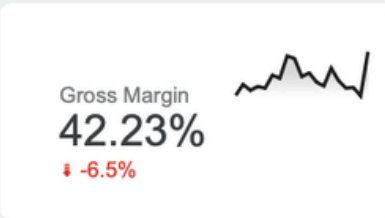
\$10.24k • \$30.81k



RECENT 3-WEEK
TRADING PERFORMANCE



Product & Inventory



Channel

Category

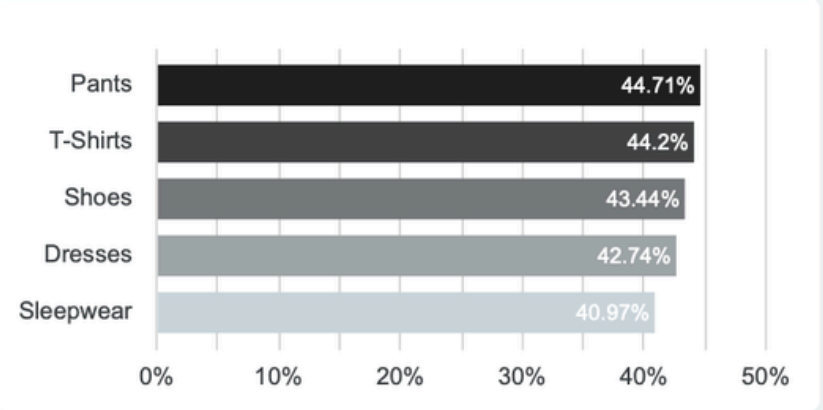
Country

12 May 2025 - 1 Jun 2025

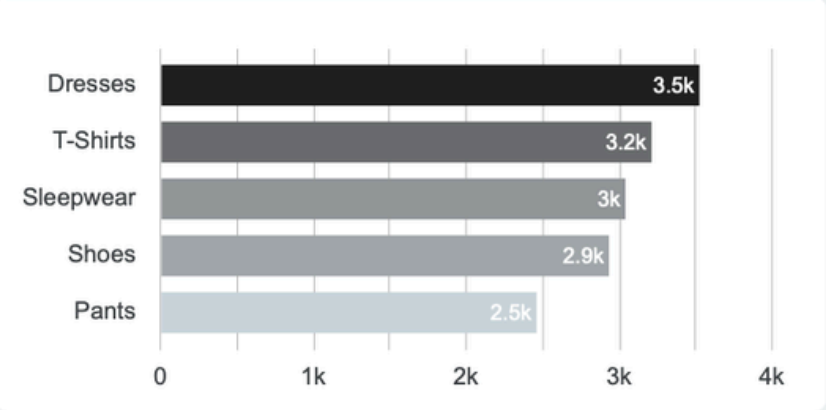
Stock Quantity
15,212

Estimated Sell-through %
30.62%

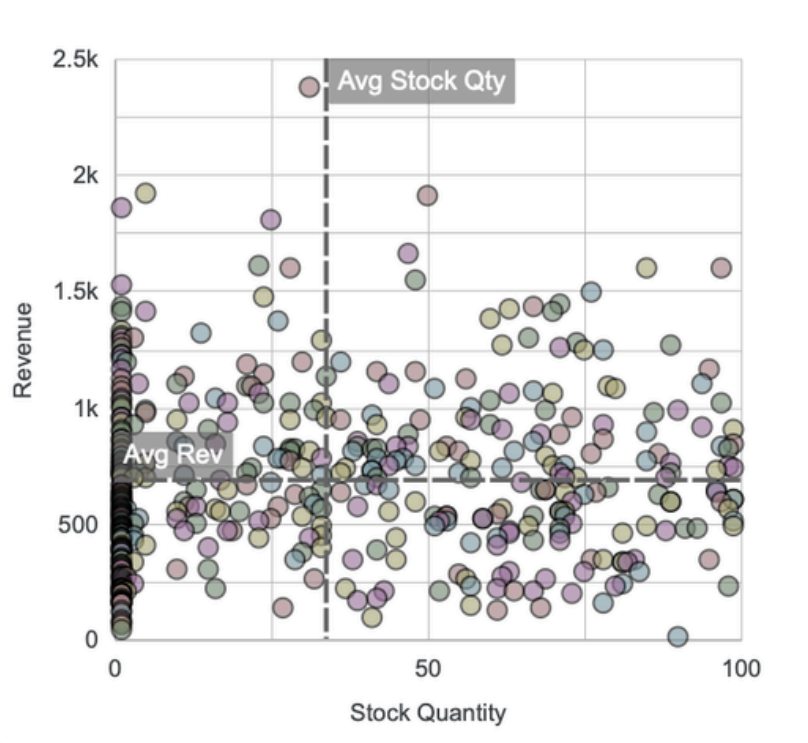
Gross Margin by Category



Total Stock Quantity by Category



Distribution of Total Stock Quantity by Revenue

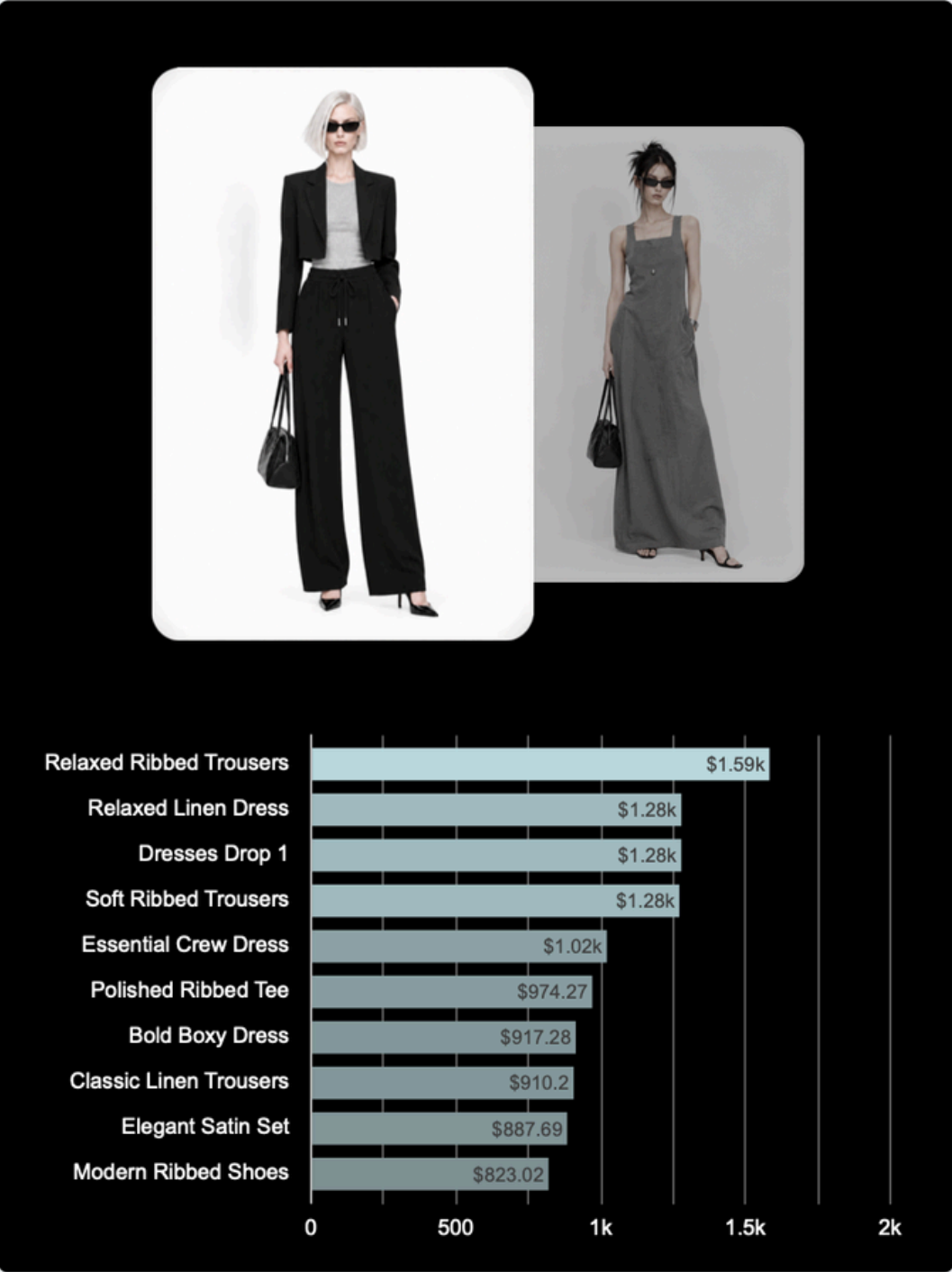


Inventory Table

	Size	Stock Quantity	Cost Price	Sell-through %
1.	XS	3,049	\$9,065.15	32.6%
2.	XL	2,354	\$6,820.07	30.33%
3.	S	2,511	\$6,592.31	30.25%
4.	M	2,330	\$5,996.36	29.22%
5.	L	2,630	\$6,581.65	28.82%
6.	40	396	\$1,591.7	36.64%
7.	38	744	\$1,913.45	26.48%
8.	36	574	\$2,043.9	34.62%
9.	35	624	\$1,640.51	32.32%

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Top 10 Products by Revenue



Products Performance Table

	Product	Revenue	Units	Cost Price	COGS	Gross Profit	Gross Margin
1.	Relaxed Ribbed Trousers	\$1,586.2	25	\$173.5	\$867.5	\$718.7	45.31%
2.	Relaxed Linen Dress	\$1,281.8	17	\$153.92	\$654.16	\$627.64	48.97%
3.	Dresses Drop 1	\$1,281.6	20	\$149.65	\$598.6	\$683	53.29%
4.	Soft Ribbed Trousers	\$1,275.2	16	\$195.24	\$780.96	\$494.24	38.76%
5.	Essential Crew Dress	\$1,024.79	15	\$141.12	\$529.2	\$495.59	48.36%
6.	Polished Ribbed Tee	\$974.27	17	\$136.52	\$580.21	\$394.06	40.45%
7.	Bold Boxy Dress	\$917.28	13	\$176.08	\$572.26	\$345.02	37.61%

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RECENT 3-WEEK
TRADING PERFORMANCE



Campaign Performance

Discount

Mkt Channel

Campaign

Country

Revenue
\$214,788

Campaign Spend
\$17,795

ROAS
12.07

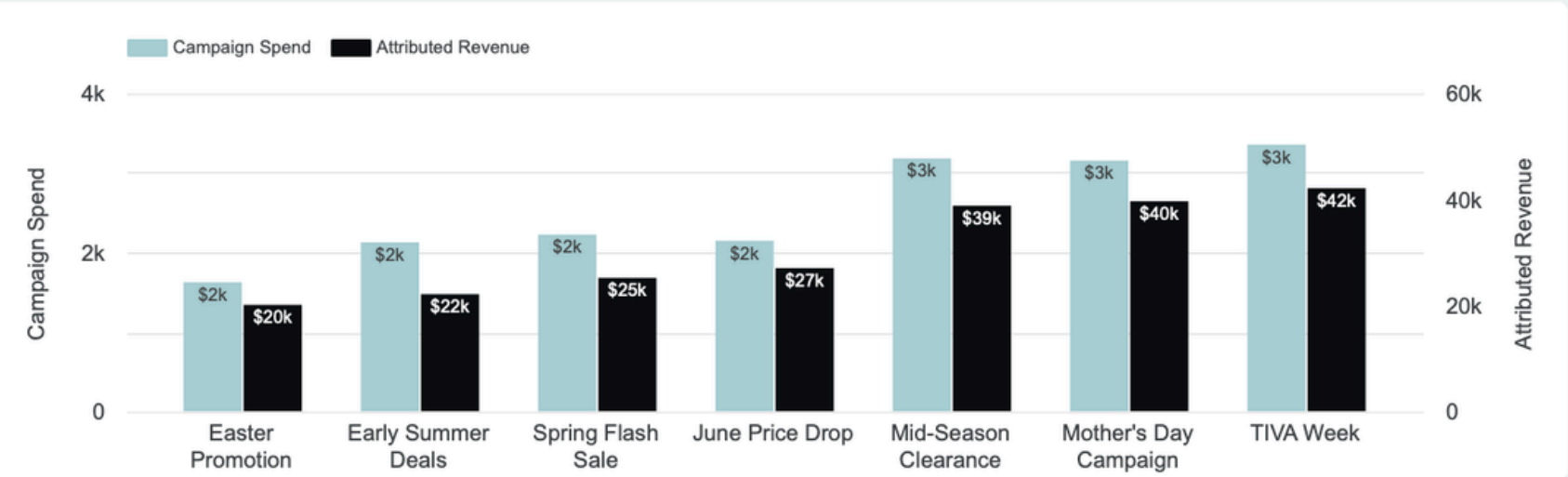
CAC
\$41.29

Conversion Rate
4.58%

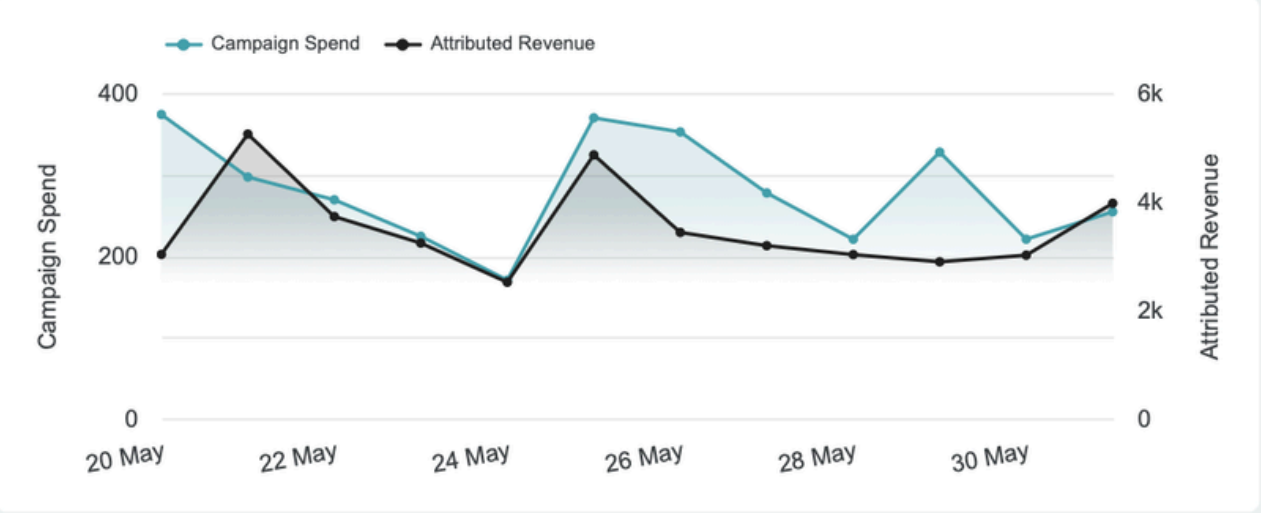
New Customers
431

AOV
\$357

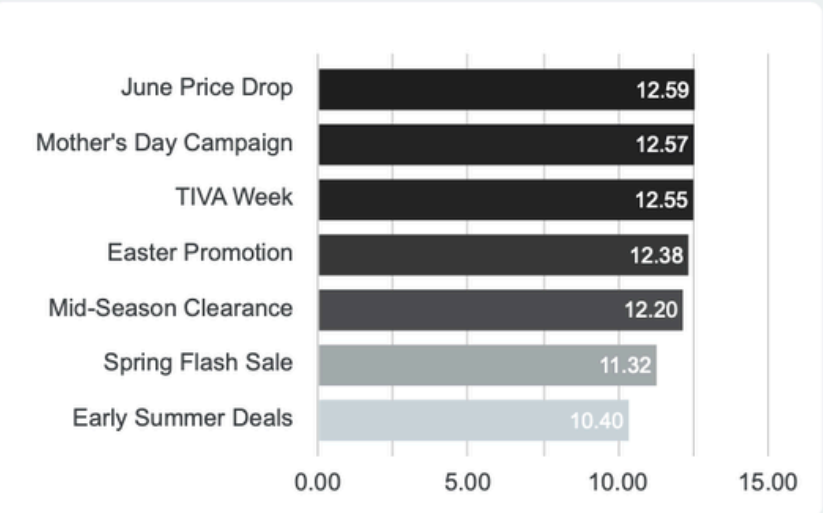
Campaign Spend vs Attributed Revenue by Campaign



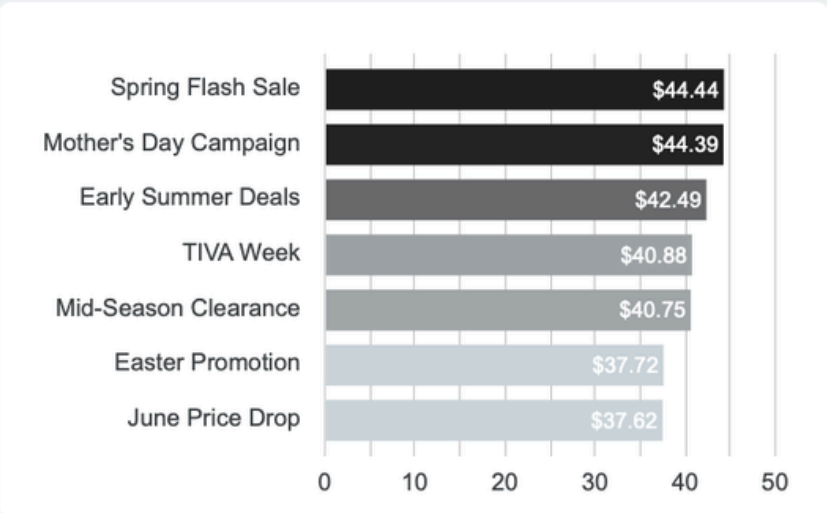
Revenue Trend - Tiva Week Campaign



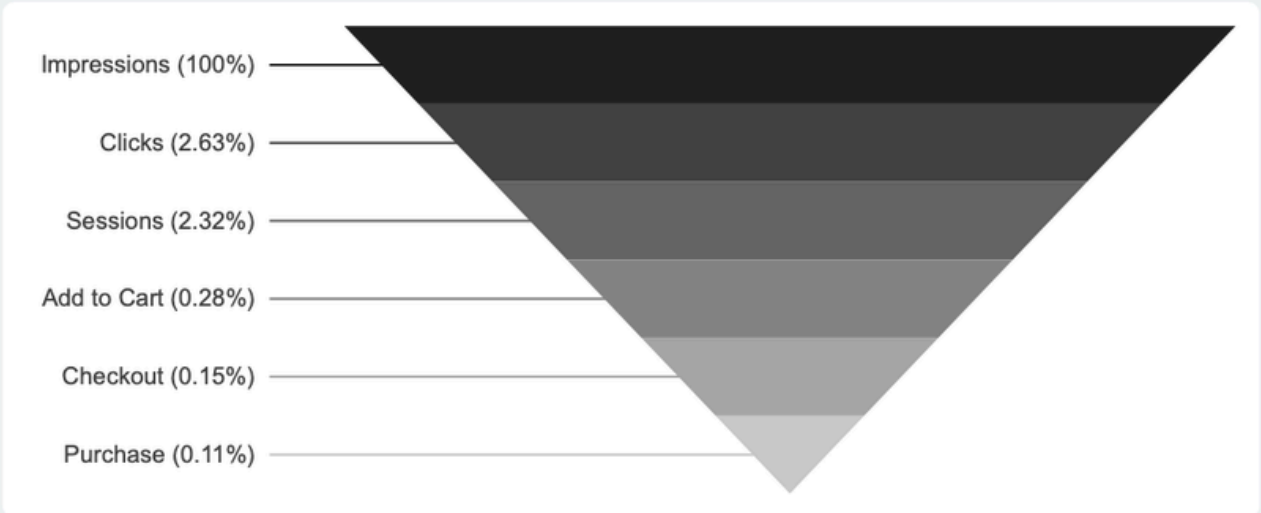
ROAS by Campaign



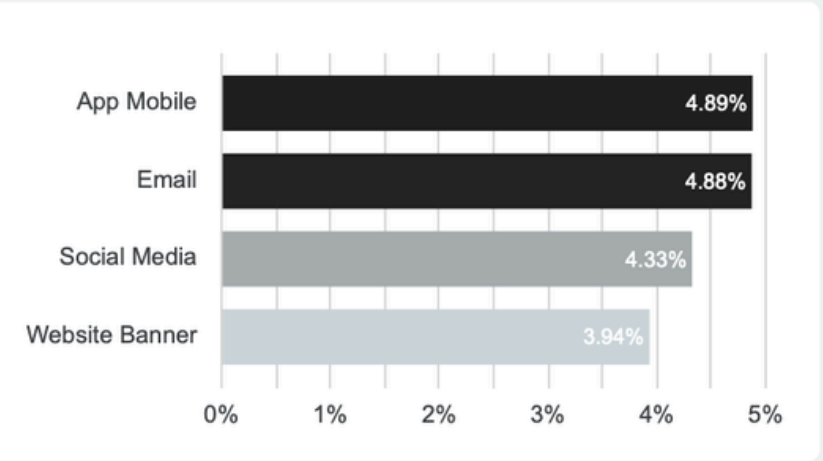
CAC by Campaign



Marketing Funnel



Conversion Rate by Marketing Channel



Campaign Performance Table

	Campaign	Country	Channel	Discount	Spend	Revenue	ROAS	CAC	CTR	CVR	New Cust
1.	Mother's Day Campaign	Italy	Email	25%	\$358.53	\$5,867.93	16.37	\$32.59	3.06%	5.21%	11
2.	June Price Drop	France	Website Banner	10%	\$378.99	\$5,822.14	15.36	\$37.9	2.36%	4.44%	10
3.	Early Summer Deals	Spain	Email	12%	\$192.36	\$2,936.26	15.26	\$24.05	2.97%	4.47%	8
4.	Easter Promotion	Italy	Social Media	10%	\$426.85	\$6,411.7	15.02	\$38.8	2.25%	4.35%	11
5.	June Price Drop	Portugal	Website Banner	10%	\$301.02	\$4,382.67	14.56	\$33.45	2.07%	4.63%	9

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CAMPAIGN-PERIOD
SPECIFIC



Customer & Market

Channel

Country

Category

Purchasing Customers
580

New Customers
341

Returning Customers
239

Repeat Purchase Rate
41%

Average Customer Value
\$559.03

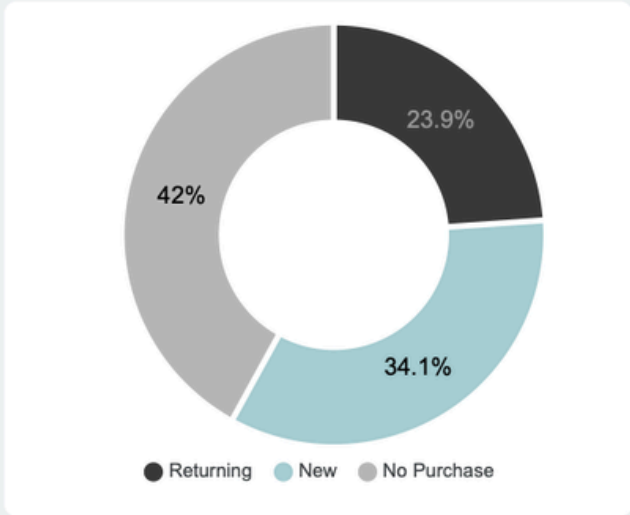
Purchase Frequency
1.56

AOV
\$358

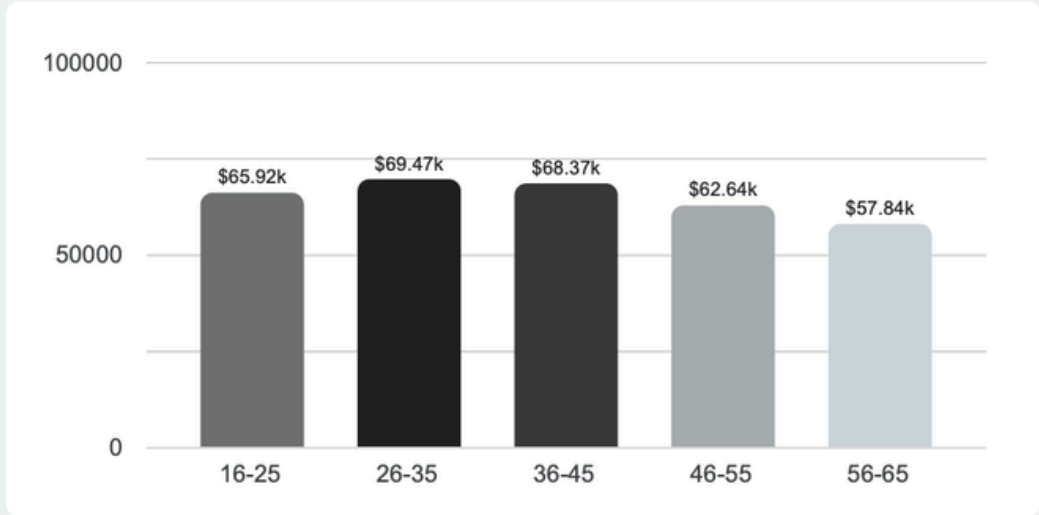
Revenue by RFM Segment and Type



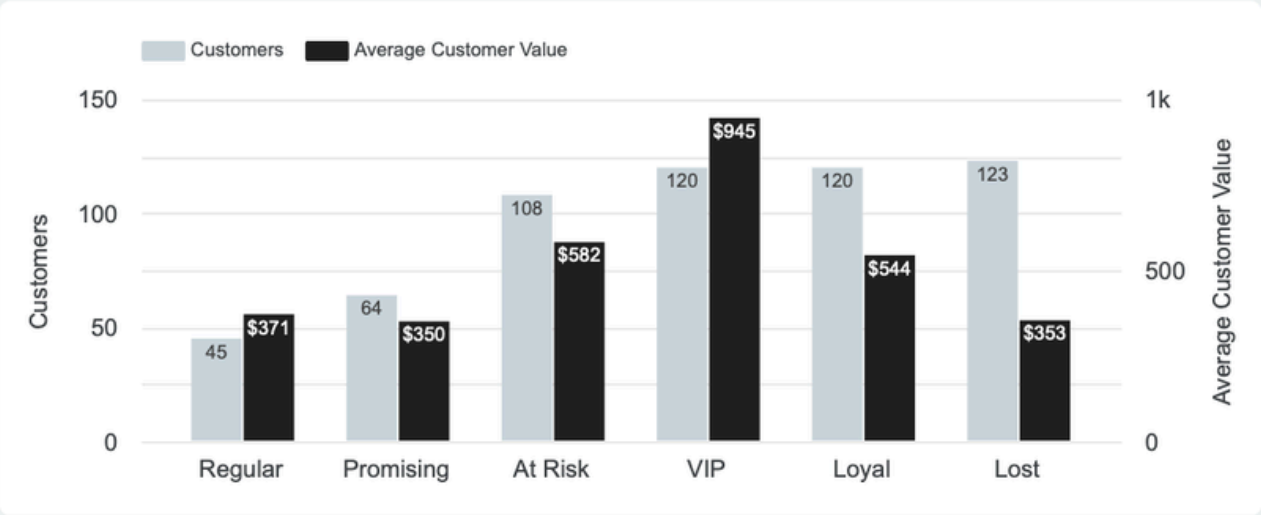
Total Customers by Type



Revenue by Age Range



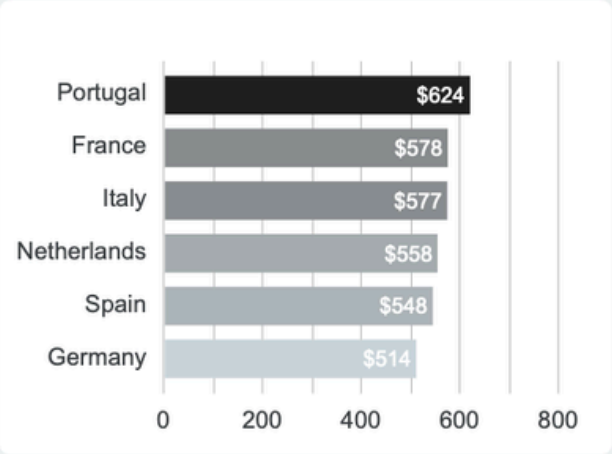
Total Customers vs Avg Customer Value by RFM Segment



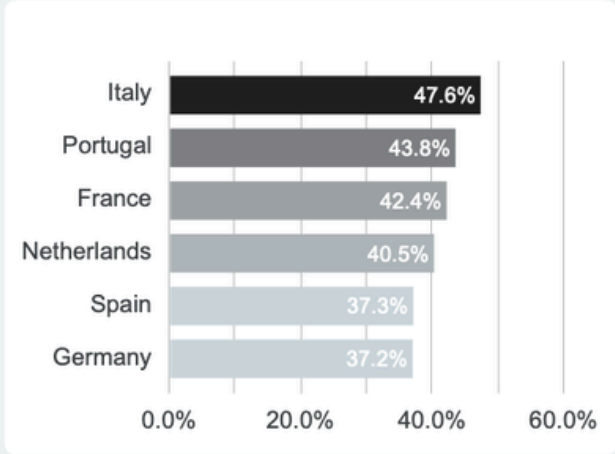
Top Customer Profiles Table

	Customer ID	Country	Age Range	Recency	RFM Segment	Revenue	Orders
1.	99	Spain	36-45	53.0	At Risk	\$2,678.59	7
2.	872	Netherlands	26-35	24.0	VIP	\$2,221.05	5
3.	220	Germany	26-35	13.0	VIP	\$2,180.51	4
4.	816	Netherlands	16-25	18.0	VIP	\$1,921.08	4
5.	253	France	26-35	4.0	VIP	\$1,882.24	4
6.	277	Italy	16-25	13.0	VIP	\$1,825.27	4
7.	776	Portugal	56-65	19.0	VIP	\$1,822.66	4
8.	190	Italy	26-35	30.0	Loyal	\$1,722.06	5
9.	564	Italy	26-35	22.0	VIP	\$1,702.7	3
10.	453	France	16-25	3.0	VIP	\$1,665.51	3
11.	346	Italy	56-65	23.0	VIP	\$1,660.56	4
12.	682	Portugal	16-25	2.0	VIP	\$1,602.07	4
13.	958	Germany	56-65	18.0	VIP	\$1,556.17	6
14.	613	Netherlands	16-25	1.0	VIP	\$1,528.54	4

Avg Customer Value by Country



Repeat Purchase Rate by Country



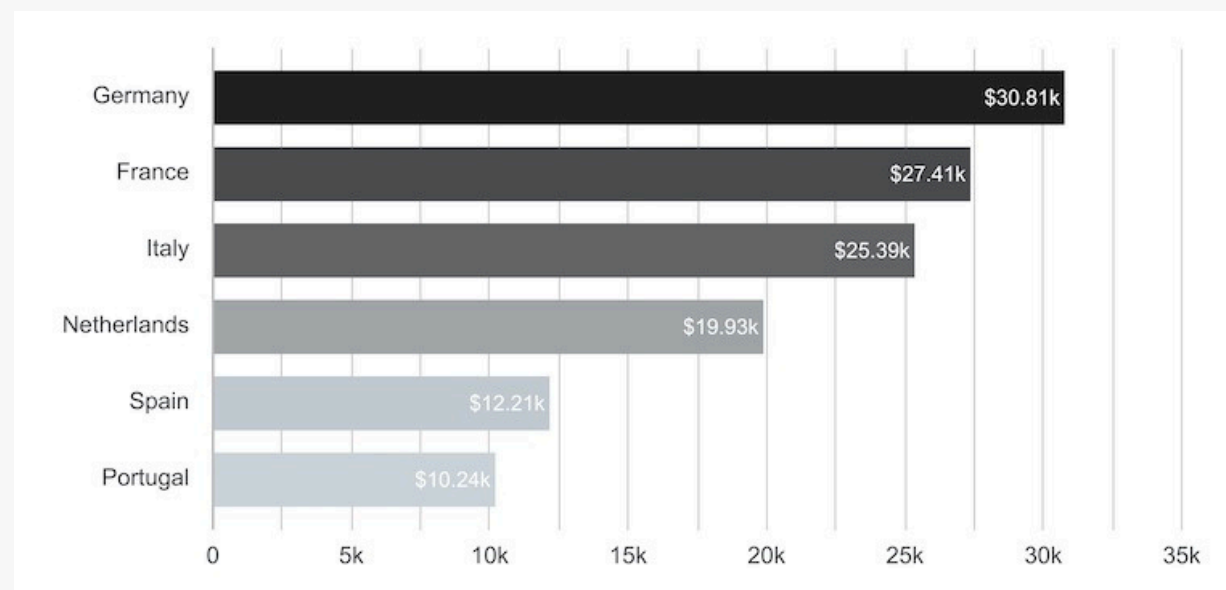
ALL-TIME CUSTOMER SUMMARY

INSIGHTS

SALES PERFORMANCE

GROWTH IS STRONG - BUT MARGIN IS SLIPPING

- Revenue is up +55% and orders +59% — but average order value fell **-2.5%** and margin dropped **-6.5%**
- When units grow faster than revenue and both AOV and margin fall, that's the signature of discount-led growth — brand is selling more, but cheaper, and giving up margin to do it
- Geographically, Germany is the biggest market at nearly \$31k, three times Portugal at \$10k



INSIGHTS

PRODUCT & INVENTORY

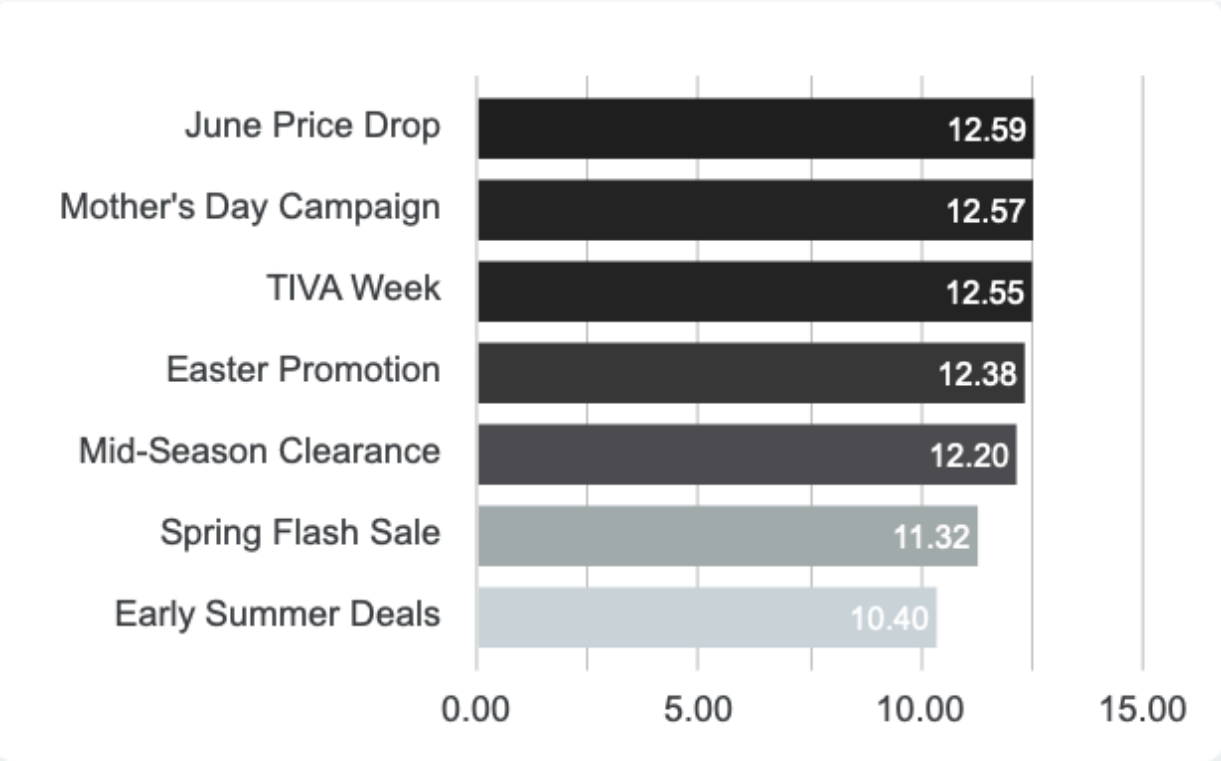
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BRAND IS SITTING ON OVERSTOCK

- Estimated sell-through is **30.6%**. For context, a healthy fashion sell-through sits between 60 and 80%, and anything **below 40%** is the industry's overstock warning line (Toolio and Shopify)
- Medium and Large were **over-bought**, all selling through in the high 20s to low 30s. Size 38 is the worst at 26%. Brand bought too deep in the middle of the size curve
- Pants earns the highest margin (44.7%) and lowest revenue (\$24k) precisely because it might be least-discounted, least-overstocked category

INSIGHTS

CAMPAIGN PERFORMANCE



DEEPER DISCOUNTS DON'T PERFORM BETTER THAN SMALL ONES

- ROAS by Campaign: June Price Drop (10% off) is the BEST at 12.59 and its CAC is the lowest at \$37.62, while the deep-discount campaigns — TIVA Week and Mid-Season Clearance (both 30% off) — sit at 12.55 / 12.20. **Deep discounts do not win** on efficiency
- Marketing Funnel: Sessions 2.32% → Add to Cart 0.28% — an **~88% drop**

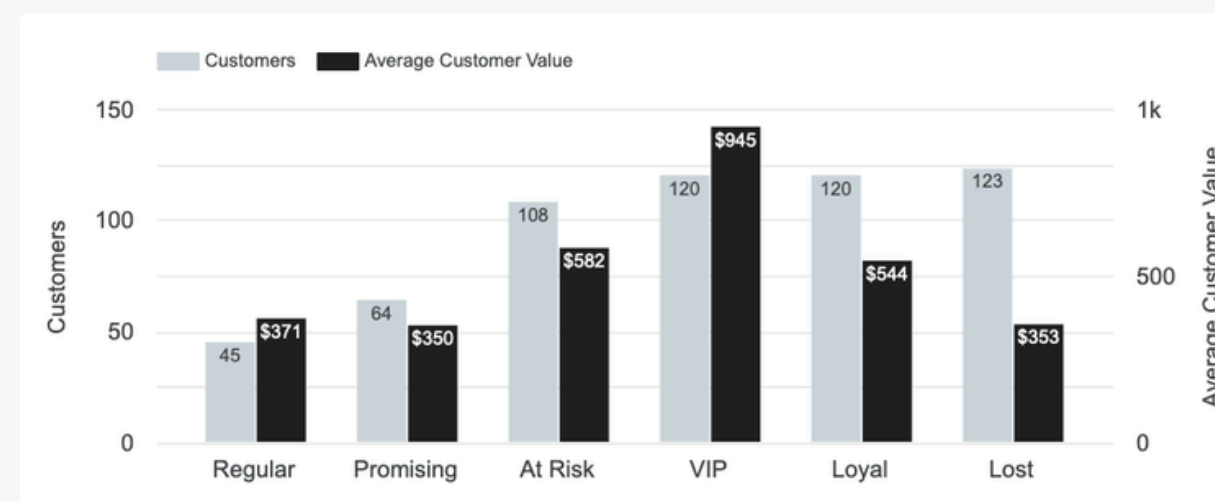
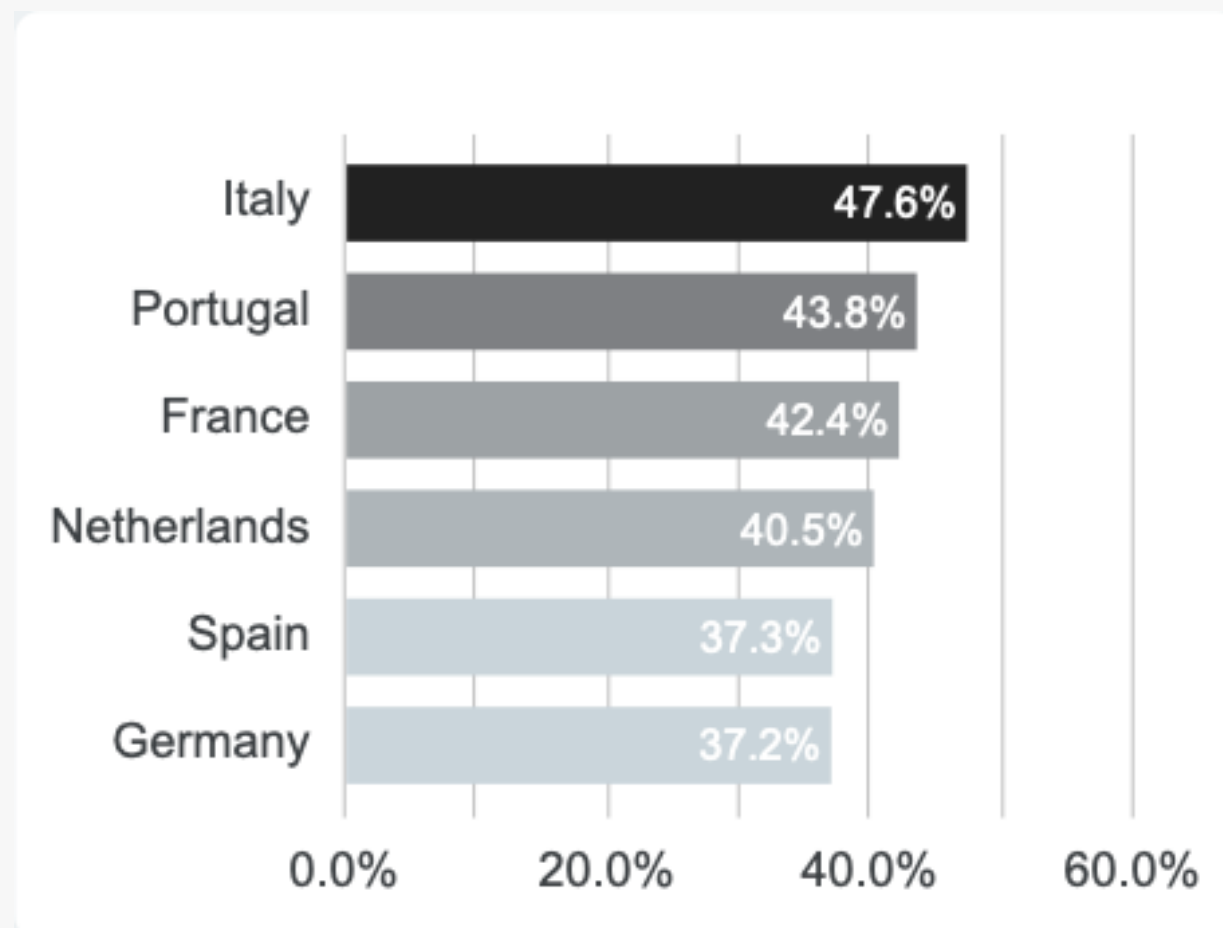
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5.	June Price Drop	Portugal	Website Banner	10%	\$301.02	\$4,382.67	14.56	\$33.45

INSIGHTS

CUSTOMER & MARKET

A THIRD OF REVENUE SITS IN CUSTOMERS WHO ARE SLIPPING AWAY

- RFM value concentration: VIP = 120 customers, \$945 avg value, \$113.45k revenue (~35% of all revenue from ~21% of buyers). **At-Risk 108 (\$582) + Lost 123 (\$353) = ~\$106k, about a third of revenue**
- At-Risk group is our second most valuable segment, so it's high-value customers who are drifting
- Loyalty is wildly uneven by market: Italy retains at nearly 48%, while Germany — our biggest market — is the worst at 37% and has the lowest customer value



STRATEGIC RECOMMENDATIONS

DISCIPLINE THE DISCOUNT

Cap standard promotions at 10–15%; reserve 25–30% strictly for genuine end-of-life clearance. Redirect the saved margin into value-adds — **free-shipping thresholds, gift-with-purchase, early access to drops** — which lift orders without training customers to wait for sales. Track a discount-dependency ratio (discounted revenue ÷ total) and hold it under 25%

REBALANCE MARKET INVESTMENT

Run a two-track market strategy. In Germany, stop pouring acquisition discounts into a leaky bucket — pivot budget to retention (second-purchase triggers, post-purchase flows). In Portugal, Italy and the Netherlands, increase acquisition — these markets turn buyers into repeat, high-value customers, so the CAC is justified.

STAND UP A RETENTION & LOYALTY ENGINE

- (a) A structured 3-email win-back sequence for At-Risk/Lost.
- (b) A tiered VIP/loyalty program — points, early access, member pricing instead of blanket discounts.
- (c) A second-purchase nudge for the 34% who are New, since first-to-second order is the hardest, highest-leverage step

WHAT'S NEXT

WHAT'S NEXT

25%

MAX DISCOUNT
DEPTH

45%

TARGET MARGIN

60%

SELL-THROUGH GOAL

48%

REPEAT RATE GOAL

WHAT'S NEXT

WHAT'S NEXT

WHAT'S NEXT

WHAT'S NEXT

WHAT'S NEXT

WHAT'S NEXT

WHAT'S NEXT

SKILLS DEMONSTRATED

PYTHON (PANDAS)

cleaned, validated, joined, and transformed multi-table fashion commerce data

DATA MODELLING

created analysis-ready datasets for sales, product, customer, inventory, and campaign analysis

KPI DESIGN

Revenue, GMV, AOV, ASP, Gross Margin, Sell-through, ROAS, CAC, CTR, Conversion Rate

CUSTOMER ANALYTICS

built RFM segmentation, repeat rate, recency, purchase frequency, and customer value metrics

CAMPAIGN ANALYTICS

created a synthetic marketing extension for spend, funnel, ROAS, and CAC analysis

LOOKER STUDIO

developed a 4-page interactive BI dashboard with filters, trends, comparisons, and decision-focused visuals

BUSINESS ANALYSIS

identified product, market, campaign, and customer opportunities while accounting for data limitations

THANK
YOU
THANK
YOU

GET IN TOUCH WITH ME

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